

NLP Coursework : PCL Detection

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https://gitlab.doc.ic.ac.uk/sa3025/nlp_coursework.git

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1 Exercise 1: PCL Paper Review

Primary Contributions

The paper introduces a novel “Don’t Patronize me” dataset which contains 10,637 annotated dataset to capture subtle condensing tones across various vulnerable communities. Unlike previous research which mostly worked on extreme and direct nature of harmful language, this data tries to capture subtle remarks directed towards underprivileged groups. The authors utilize an expert-led annotation process to address subjectivity, resulting in a unique taxonomy consisting of high-level categories (e.g., The Saviour, The Poet) and seven fine-grained sub-categories.

Furthermore, the authors establish a rigorous baseline for PCL detection using various architectures showcasing that subtle tones of PCL is a more challenging task.

Technical Strengths

The paper formalizes PCL detection as an NLP problem and tackles it with a systematic approach.

Rich Dataset

The dataset was annotated by three experts: two primary annotators ranked the text, while a third acted as a referee to resolve disagreements. These scores are combined to a 5 point scale and capture borderline cases as well. Additionally, the dataset is highly diverse, sourced from news outlets across 20 English-speaking countries and covering 10 keywords related to vulnerable groups.

Modeling

The authors assessed a diverse range of architectures for both binary and multi-class Patronizing and Condensing Language (PCL) classification tasks. They experimented with different types of models, such as traditional (SVM-WV and SVM-BOW), neural networks

(BiLSTM), and more advanced architectures such as the language models (BERT, RoBERTa, and DistilBERT). To ensure robust results, the authors conducted 10-fold cross-validation and hyperparameter tuning with early stopping.

Key Weaknesses

There are few key weaknesses present in that paper with some being mentioned by authors without providing technical resolutions.

Low Inter-Annotator Agreement

The Kappa Inter-Annotator Agreement between the two annotators are low with only 41% agreement which was later increased to 61% by removing edge cases. This low score highlights the subjective nature of PCL such that even experts struggle to reach a consensus.

Need for World Knowledge

The authors have mentioned the need for external world knowledge in multiple instances and even advanced models like BERT/RoBERTa struggle in these areas. Categories such as “Shallow solutions” or “Presuppositions” requires non-trivial knowledge which makes it harder to predict them accurately. Even though the authors identify these issues, they do not propose any technical means to incorporate them making it a very challenging issue to model them.

2 Exercise 2: Exploratory Data Analysis (EDA)

EDA 1: Statistical Profiling

Class and Token Distributions

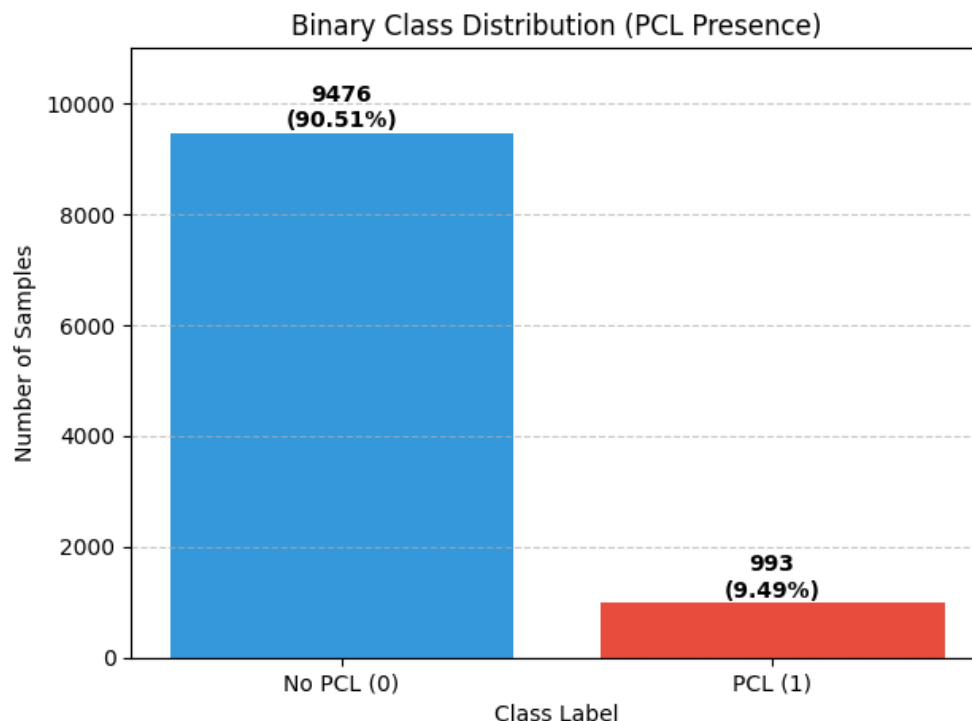


Figure 1: Binary Class Distribution of the Don't Patronize Me! Dataset.

Analysis:

A basic statistical profiling of the dataset revealed that there is a severe class imbalance, with approximately 90% of the sentences not condescending in nature and only 10% containing actual patronizing language. If it was trained directly, the model would likely predict majority of the class for all input and still achieve an accuracy of 90%.

Impact:

To allow model to learn the minority class well, I will implement PyTorch "WeightedRandomSampler" which will ensure that the model receives a balanced 50/50 of each class per batch on average. As mentioned above, using accuracy will produce incorrect results and therefore I chose to use Precision, Recall and f1 score as the evaluation metrics.

Metric	Value(tokens)
Average Length	48
Minimum Length	1
Maximum Length	909
25% Quantile	30
Median Length (50%)	42
75% Quantile	60
99% Quantile	159
Total Unique Tokens (Vocab Size)	33,752

Table 1: Statistical Profiling of Dataset Token Counts and Vocabulary

Analysis:

From the above table ,the token distribution is right-skewed with the average length of input is 48 tokens and 99% of the length between 8 and 159 tokens. Less than 1% of data has token size above 150, which indicates its safe to assume tokenizer’s max length of the parameters to be 256 tokens without significant data loss. Even though the maximum token length is 909 , the standard RoBERTa-Large architectures are limited to a maximum sequence length of 512 and therefore will get truncated. As well, there are 33,752 unique tokens across the dataset comprising of the vocabulary.

Impact:

The size of attention layer depends on the size of the input and the architecture scales quadratically in both memory and compute $O(N^2)$.By limiting the input size, we can avoid the CUDA out of memory failures. Based on the above statistics, it is safe to truncate data to 256 tokens by setting the tokenizer’s max length. From an architectural standpoint, the size of vocabulary will dictate the capacity requirements of the embedding layer.

Word Clouds & Frequency



The word cloud highlights a high frequency of verbs like “said,” suggesting the data is sourced from news articles and journalistic reports, which aligns perfectly with our data sources. Furthermore, words such as “women”, “immigrants”, “children”, and “refugee” clearly demonstrate the focus on vulnerable demographics. Among the top 100 words, explicit condescending terms are quite rare, showcasing the subtle nature of the Patronizing and Condescending Language (PCL) within the dataset. Additionally, the dataset includes valuable metadata, such as the 10 specific keywords and demographic tags for each entry. Finally, about 40% of the text consists of stop words, which is exactly what we would expect from sentences of a journalistic nature.

Due to the high frequency words such as “homeless” and “poor families”, there is a significant risk that the model might simply associate those terms with PCL and take shortcuts in learning. To tackle this, I will append the keyword and geography metadata to the main text, using the `< /s >` separator to enable effective tokenization by RoBERTa. By exposing this information, the model will be pushed to look beyond superficial keywords and learn deeper semantic meanings. Additionally, in cases where there is ambiguity about “who” the text is referring to, keywords like immigrant or refugee will provide crucial context. Finally, given the subtle nature of the data, I chose to use RoBERTa-large with custom add-ons, retaining all stop words to preserve the necessary grammatical structure.

3 Exercise 3: Proposed Approach & Rationale

Proposed Approach

The proposed approach for PCL detection consists of the following stages:

- **Data Preprocessing:** The dataset will be cleaned to remove artifacts, and metadata (including keywords and geographical data) will be appended to the primary input text. These elements will be segregated using the `</s>` token as a delimiter to align with the model’s pre-training (e.g., `[keyword] </s> [geography] </s> [Text]`).
- **Batch Sampling:** To mitigate the 90/10 class imbalance, a PyTorch `WeightedRandomSampler` will be implemented. This ensures a 50/50 class distribution within each training batch to prevent majority-class bias.
- **Multi-Head Architecture with Dropout:** The RoBERTa-Large model will be fine-tuned using a dual-head architecture. One linear head will address the binary classification task (PCL vs. non-PCL), while a second head will simultaneously perform multi-class categorization. A dropout rate of 0.3 will be applied before the linear heads.
- **Hyperparameters and Training:** Fine-tuning will utilize a learning rate of 1×10^{-5} over a maximum of 3 epochs. A model checkpointing strategy will be employed to save weights only when there is strict improvement in the F1-score.
- **Post-Training Optimization:** The classification threshold will be tuned based on validation set probabilities to refine the decision boundary and optimize performance for imbalanced data.

Rationale and Expected Outcomes

- **Metadata Injection:** High-frequency words like “homeless” often trigger lazy learning, where the model ignores context. Adding meta-information forces the model to look beyond keywords, while the `</s>` token ensures the tokenizer clearly segregates this information.
- **Weighted Sampling:** Enforcing a 50/50 split ensures the loss function penalizes errors in both classes equally and preventing larger majority class predictions .
- **Architectural Efficiency and Dropout:** Simultaneous training on two tasks forces the model to learn richer, more nuanced internal representations. The 0.3 dropout rate will prevent model from becoming overly dependent on specific neurons and generalize better.
- **Stability and Generalization:** A small learning rate ensures stable convergence. Given the limited size of the training data, restricting training to 3 epochs and using strict F1-based checkpoints prevents RoBERTa-Large from overfitting.
- **Threshold Calibration:** Post-training thresholding finetunes the decision boundary to mitigate false positives introduced by weighted sampling, ensuring alignment with the true data distribution.

4 Exercise 5: Global and Local Evaluations

Error Analysis

Paragraph	Actual	Model	Baseline
Camfed would like to see this trend reversed. It would like to see more girls in school. Basic Education Statistics in Tanzania (BEST 2010) show that only 18 percent of girls have completed secondary school education. This is why Camfed supports girls from poor families to obtain secondary education and its efforts have seen many go to university.	1	0	0
The World Health Organization did not give a reason for the increase in deaths, but a provincial health official in Sindh said that the disease hit areas where poor families did not vaccinate their children.	1	0	0
I am a Jaffna Tamil residing in the USA and I see the Sri Lankan professionals over here, both the Tamils and Sinhalese, holding top positions in Research, Universities HighTech industries, Law, and Finance. But then looking at the sterile Arab countries all we see are Sri Lankan women working as maids and servants to the rags to riches Arabs. A self respecting society wouldn't allow its women to be transported as maids to foreign land.	0	1	0
As leaders, we will personally support victims but we ask the government to also help. We are going to clear all victims' hospital bills and we want to ensure that those left homeless receive shelter.	0	1	0
President Muhammadu Buhari is on a divine mission to rescue the Nigeria from the brink of collapse and hopelessness, Speaker of the House of Representatives, Rt. Hon. Yakubu Dogara has said.	1	1	0
McAfee said he was helping the locals by feeding poor families and providing them with jobs.	1	1	0
Fasting bridges the gap between rich and poor , sustained and impoverished , fulfilled and needy . This experience should then inspire compassion and mercy , which is manifest by generosity of wealth and time to help those in need .	1	1	1
These lengthy , inefficient and burdensome processes result in more hindrance than assistance to SMEs that are more in need of government assistance than larger firms to enter international markets , "" the study highlighted .	0	0	0

Table 2: Comparative Analysis of Actual, Custom Model Predictions & Baseline Predictions

The divergence in performance of the baseline RoBERTa-base model and the custom RoBERTa-large model reveals that the PCL detection is as dependent on the pragmatic context as it is on the semantic understanding. Below are some detailed analysis of few of the examples and the difference in performances across the models.

Shared Failures:

This is the case where both the custom and the baseline models fail to classify the context correctly. From the first 2 above examples in the table, we see the models failing to capture tones embedded in more a formal, institutional way. For example, the use of institutional bodies such as WHO , CAMFED can misdirect the models into more of statistic over patronizing message.

Custom Model Failures:

Custom model is facing issues of being over-sensitive at times even when the text is neutral in nature. This could primarily be due to the injection of metadata making the model more aware of the keywords. Whereas in the case of baseline model, the simplistic nature of it helps it avoid this and stay more neutral. However, in the case of Srilankan example, it lacks the social context to realize that an insider performing a self-critique or a leader asking for help is often not being patronizing.

Custom Model Success:

Though the baseline’s simplicity can be of advantage, the custom model has outperformed in identifying complex linguistic tones, especially the poetic tone effectively. For example, these use poetic phrases such as “divine missions” which reflect the unbalanced power dynamics, while the baseline is misled by the positive vocabulary (rescue, feeding, jobs). The custom model’s deeper attention mechanism allows to capture these nuances effectively, moving beyond simple sentiment analysis.

Finally, the error analysis reveals the nature of PCL detection as one of the most technically challenging tasks in NLP due to its subjective and subtle nature. This is also seen from the low agreements of patronising language between the annotators. Many samples exists on margins of condescending making it difficult for experts let alone machine learning models. But the results show that the custom model does a fairly good job in capturing these nuances and interpreting complex linguistic context effectively.

Confusion Matrix

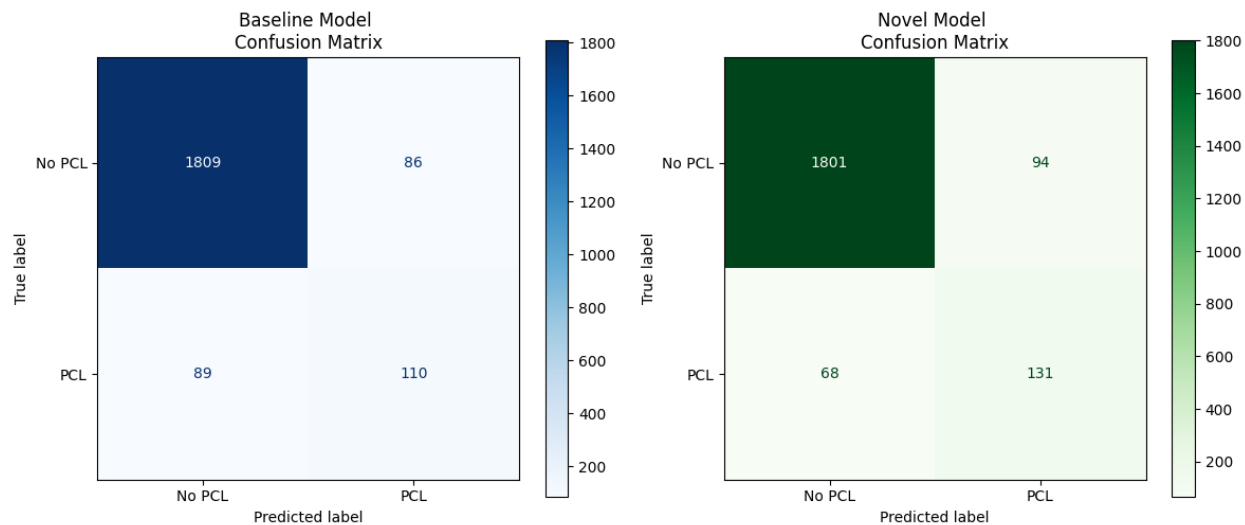


Figure 3: Confusion Matrix Baseline Model vs Custom Model

The task of the model is to detect the presence of patronising language which is the minority class in dataset. Due to the high imbalance of data, models generally struggle to detect PCL and predict majority class safely to achieve higher metrics. This is seen here in the baseline confusion matrix where the model only predicts 110 vs 131 PCL cases compared to the novel approach. We see nearly 10% jump in the recall(sensitivity) which captures more than two-third of the actual PCL class(199 samples). We see a minimal impact on the No PCL class with just loss of 8 negligible given the high volume of the data. Overall F1 scores shows an improvement from 0.55 of base model to 0.617 on the custom model which is considered as a significant improvement.

Precision-Recall Curve

To go beyond standard accuracy metrics, a Precision-Recall (PR) Curve was generated to evaluate the model’s performance across all possible classification thresholds. IT captures the trade-offs between catching subtle condescension (Recall) and avoiding the mislabeling of neutral information (Precision). The average precision(AP) score is the weighted mean of precision at every threshold and indicates the area under the PR curve. The achieved Average Precision (AP) of 0.61 demonstrates the model’s ability to rank positive instances effectively. The red marker represents the threshold (0.89) that was chosen from the post-training threshold tuning on validation set. Models such as Roberta-Large are known to be overconfident in prediction of PCL (Almendros et al., 2020). as well as the use of weighted random sampler artificially increases the positive cases during training. This can lead to inflation in probabilities which are corrected by stricter threshold.

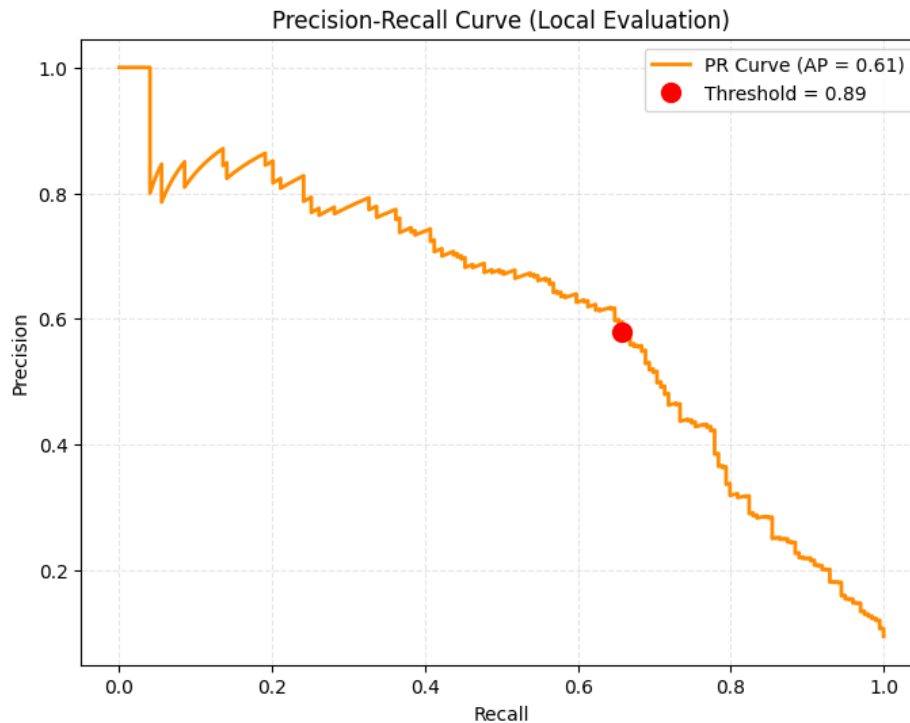


Figure 4: Precision-Recall Curve of Custom Model

References

Carla Perez Almendros, Luis Espinosa Anke, and Steven Schockaert. 2020. *Don't Patronize Me! An Annotated Dataset with Patronizing and Condescending Language towards Vulnerable Communities*. In Proceedings of the 28th International Conference on Computational Linguistics, pages 5891–5902, Barcelona, Spain. International Committee on Computational Linguistics.